DIY CONCRETE

GARDEN SET

Planters · BirdAth · Curved Path — One Weekend Build

Step-by-Step Professional Guide

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ABOUT THIS PROJECT

This guide shows you how to build a complete concrete garden set from scratch — square raised planters, a geometric octagonal bird bath on a pedestal, and a sweeping curved concrete path. Everything is made using basic formwork built from timber and melamine board. No specialist equipment needed — just concrete mix, a wheelbarrow, and one weekend. The finished result looks like a professional landscaping job worth thousands.

Difficulty	Intermediate
Time	Weekend (2 days)
Cost	$80–$150 total
Result	2 planters + bird bath + curved path
Durability	Permanent, frost-resistant

MATERIALS & TOOLS

ITEM	QTY	NOTES
Concrete mix (C20/25)	200 kg	Bagged or bulk; standard structural mix
Timber formwork (melamine board)	10 sheets	For planter and bird bath molds
Screws & timber battens	1 pack	To assemble formwork panels
Wheelbarrow & mixing tools	1 set	Or electric mixer for larger batches
Work gloves & safety glasses	1 pair each	Concrete is highly alkaline
Trowel & float	1 set	For leveling and smoothing surfaces
Vibrating tool or rod	1	Removes air pockets from concrete
Waterproof membrane/liner	1 roll	For planter drainage layer
Drainage gravel	2 bags	Layer at bottom of each planter
Angle grinder + diamond disc	1	Optional — for cutting formwork precisely
Concrete sealer	1 can	Protects finished surfaces from frost & stain

STEP-BY-STEP PROCESS

01 PLAN & MARK YOUR LAYOUT

Plan and mark the garden layout on the ground

Mark the position of your curved path, planters and bird bath with spray paint or sand. The curved path is the anchor — everything else positions off it. Use a garden hose to lay out the curve first, then mark it permanently. Plan the path to be at least 60 cm wide for comfort.

★ PRO TIP A curved path with a constant radius looks more elegant than a free stake and string to draw it.

02 MIX CONCRETE

Mixing concrete in the wheelbarrow by hand

Add concrete mix to the wheelbarrow and mix by hand or with a mixer. The mix should be firm and hold its shape when squeezed — not too wet. For planters, use a slightly stiffer mix (less water) to hold sharp edges. For the path, a slightly wetter mix is easier to level.

★ PRO TIP Mix in small batches if working alone. Concrete starts to set in 30–4 more than you can place.

03 BUILD SQUARE PLANTER FORMWORK

Assembling timber formwork for the square planter

Cut melamine board panels to the planter dimensions (e.g. 40×40 cm exterior, 30×30 interior). Screw together using timber battens on the outside. The inner box (creates the hollow) must be suspended centered inside the outer box. Apply oil or release agent to all inner surfaces of the formwork.

★ PRO TIP Melamine board releases cleanly from concrete with no sticking and face. Use it every time.

04 POUR & VIBRATE THE PLANTERS

Pouring concrete into the planter formwork

Place formwork on a flat surface. Pour concrete in layers of 10 cm, vibrating after each layer. Use a rod or rubber mallet tapped on the formwork to release air bubbles. Fill to the top and screed level. Leave the base 3 cm below the rim for drainage gravel.

★ PRO TIP Drill 3 drainage holes in the base formwork (20 mm diameter) befor concrete will fill these bolts — knock them out after demolding.

05 BUILD THE BIRD BATH PEDESTAL

Finished square pedestal base cured and demolded

The pedestal is a solid square column — use simple timber formwork, 25×25 cm, 50 cm tall. Pour concrete, vibrate well, and cure for 48 hours before demolding. The pedestal must cure fully before the basin is placed on top — it carries significant weight.

★ PRO TIP Add 4 steel rebar rods (8 mm diameter, 60 cm long) to the pedestal should protrude 10 cm from the top to anchor the basin.

06 CAST THE OCTAGONAL BASIN

The geometric octagonal bird bath basin on the pedestal

The octagonal basin is cast in a purpose-cut timber mold. Cut 8 identical trapezoidal panels from melamine board and assemble into an octagon. The inner mold creates the hollow. Pour concrete, vibrate, and smooth the top rim with a float. Cure 48 hours, then demold carefully. Place on the pedestal — the rebar anchors it in place.

★ PRO TIP Sand the rim of the basin smooth with 120-grit sandpaper before pla will be uncomfortable for birds.

07 FORM THE CURVED PATH

Laying the curved concrete path sections

Excavate 10 cm deep along the marked path line, 60 cm wide. Compact the base with a plate compactor or by hand. Add 5 cm of gravel and compact again. Use flexible timber edge formwork (thin strips) to follow the curve. Pour concrete sections of 1–1.5 m at a time, screed level, and finish with a float.

★ PRO TIP Cut expansion joints every 1.2 m with a grooving tool while the conc prevents cracking as the slab moves with temperature changes.

08 CURE & PROTECT

Finished path and planters during curing

Cover all freshly poured concrete with plastic sheeting for at least 48 hours. Mist with water twice daily for the first 3 days to slow the cure and prevent cracking. Do not walk on the path for 72 hours. Full strength is reached at 28 days.

★ PRO TIP Curing in hot weather is critical. Rapid drying causes surface crackin and damp.

09 DEMOLD & FINISH

Demolding the planter formwork after curing

Carefully unscrew and remove formwork panels. If any surface has minor voids, fill with a 1:1 cement-water paste and smooth. Sand any sharp edges with 60-grit paper. Apply a penetrating concrete sealer to all surfaces — planters, path and bird bath.

★ PRO TIP Sealer is not optional for planters. Freeze-thaw cycles will destroy u within 2–3 winters.

10 PLANT, FILL & ENJOY

Complete concrete garden set at golden hour

Add drainage gravel to the bottom of each planter (5 cm layer), then quality potting mix. Plant seasonal flowers, herbs or small shrubs. Fill the bird bath basin with clean water — birds will find it within days. Stand back and admire your work. This is a permanent landscape feature that will last decades.

★ PRO TIP Group planters in odd numbers (1, 3, 5) at varying heights for a prof
landscaping look.

PRO TIPS & TRICKS

Formwork is Everything	The quality of your concrete work is 80% formwork quality. Use melamine board, screw tightly, oil every surface. Gaps in formwork = blowouts. Over-engineer it — it is faster than rebuilding.
One Mix Ratio, Three Textures	Use the same C20/25 mix for everything — just vary the water slightly. Planters: stiff mix (less water, sharp edges). Path: medium (easy to screed). Bird bath: slightly wetter (fills thin sections).
Rebar Makes It Permanent	Add at least 4 rebar rods to any piece taller than 30 cm. Concrete is strong in compression, weak in tension — the rebar handles the tension. A planter or pedestal without rebar will crack when it freezes.
Sealer Every Season	Apply a penetrating sealer every 2 years. It takes 30 minutes and prevents all freeze-thaw damage, moss growth, and staining. This is the single most important maintenance task.
Colour the Concrete	Add iron oxide pigment (grey, charcoal, sand) to the mix for a consistent, deep colour that won't fade. Use 2–4% by weight of cement. Mix pigment into dry cement before adding water for even colour throughout.

TROUBLESHOOTING

PROBLEM	CAUSE	SOLUTION
Formwork blew out during pour	Too much water; not enough screws	Stiffer mix; more screws every 20 cm; pour in layers
Surface cracking after cure	Dried too fast; no expansion joints	Keep covered & damp 72h; cut joints every 1.2 m
Voids / honeycombing on surface	Air not vibrated out	Vibrate every layer; tap formwork with mallet
Planter leaking at base	No drainage or crack formed	Drill drainage holes; seal inside with bitumen paint
Bird bath wobbling on pedestal	Rebar not used; uneven base	Rebar anchor + mortar bed under basin
Path surface too rough	Over-trowelled when too wet	Float earlier; one pass only when sheen disappears
Concrete colour uneven	Pigment added too late or unevenly	Mix pigment into dry cement before water; weigh consistently

Thank you for choosing Made By Hand!

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